

Effects of Soil Physical Structure on Solute Transport in a Weathered Tropical Soil

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ABSTRACT

Effects of soil structure on solute transport were studied in a clayey, well-aggregated Oxic Dystropept under grass and secondary forest at La Selva, Costa Rica. Both fine pores ($<0.1\ \mu\text{m}$) and coarse pores ($>30\ \mu\text{m}$) are abundant, as indicated by a water content of $0.37\ \text{m}^3\ \text{m}^{-3}$ at $-1500\ \text{kPa}$ matric potential, the shape of the moisture release curve, and an initial infiltration rate (at field capacity) averaging $3900\ \text{mm}\ \text{hr}^{-1}$. Field application of Rhodamine B dye without ponding (simulating heavy rainfall) showed preferential flow along decayed-root channels, animal burrows, cracks, and other macropores. Dye application to intact cores under conditions of ponded steady-state flow gave a good correlation between flow rate and total stained area ($p < 0.01$ under forest; $p < 0.05$ under grass). Solute (CaCl_2) breakthrough occurred very rapidly, often after <0.1 pore volumes had percolated; however, relative concentration of the effluent did not exceed 0.95 even after five pore volumes had percolated. These results indicate that most water flows between aggregates or through macropores (as preferential or channelized flow), even when the fine pores are not fully saturated, and in effect bypasses the fine pore space. When solution inflow was interrupted after ~ 1.8 pore volumes, then resumed after 10 min pause, relative concentration of the effluent dropped by 10 to 40%, then rose again. This indicates that solute diffused slowly into the aggregates. Taken together the results suggest that this soil strongly resists leaching. Preferential water flow may serve to prevent nutrient loss from the matrix of all highly aggregated soils and of all noncultivated soils in which animal activity and turnover of woody roots create abundant macropores.

IN MANY STUDIES of solute transport, soils are assumed to be a homogeneous porous medium in which water moves downward as a coherent wetting front. Over the last decade, soil scientists have come to appreciate increasingly that two broad classes of soils deviate markedly from this idealized norm. Studies of miscible displacement, for example, show that highly aggregated soils require more pore volumes of water to leach nutrients than do nonaggregated soils (Anderson and Bouma, 1977a, b; White, 1985). The explanation is that, in aggregated soils, water flows between the aggregates without displacing the water held in fine pores within aggregates (Van Genuchten and Wierenga, 1976; Rao et al., 1982; Van Genuchten and Dalton, 1986). Even poorly aggregated soils, however, may have a pore-space arrangement that prevents water from moving downward as a front through the soil matrix. Macropores—pores that are too large to retain water by capillarity—are abundant in undisturbed soils, such as occur under forest, pasture, and perennial crops (e.g., Aubertin, 1971; Trudgill et al., 1983; Field et al., 1984; McVoy, 1985; Watson and Luxmoore, 1986). Taking the form of cracks, animal

burrows and holes left behind after coarse roots decay, such macropores create preferred paths along which water moves as channelized flow, thus bypassing the soil matrix.

Both macropore development and stable aggregation create a similar water flow regime, often called biphasic because water can flow rapidly through large pores while the water in fine pores remains relatively immobile. A biphasic flow regime is known to increase the rate of movement of fertilizers, pollutants and microorganisms through the soil by slowing their entry into the soil matrix (Thomas and Phillips, 1979; Bouma, 1981; Smith et al., 1985; White, 1985; Tindall et al., 1986). Biphasic water flow may also restrict leaching of solutes from the soil matrix. For example, biphasic flow increases the amount of water needed to desalinate soil (White, 1985). Preferential flow has been implicated as a factor restricting the leaching of nitrate (Wild, 1972) and cations (Nortcliff and Thornes, 1978) in tropical soils.

Our objective in this study was to measure the importance of the biphasic flow regime in controlling solute transport through a highly weathered, strongly aggregated Oxic Dystropept at La Selva, Costa Rica. The work forms part of a larger study of nutrient mobility in La Selva soils in which we have measured effects of microbial processes on soil pH and charge-status (Sollins et al., in press), and rates of water infiltration (Radulovich and Sollins, 1985) and drainage (Radulovich and Sollins, 1987).

MATERIALS AND METHODS

Site Characteristics

The study area is located near Puerto Viejo in northeastern Costa Rica ($10^\circ 26'\text{N}$, $83^\circ 59'\text{W}$) at about 50 m elevation in the La Guaria Annex of the La Selva Biological Stn. Annual precipitation averages 4000 mm, and mean annual temperature is 24°C (La Selva Meteorological Stn, 1957–83). The site is on an old, gently undulating river terrace about 15 m above the Rio Sarapiquí.

Two sites with different vegetation cover were studied, both cleared of primary forest in the 1960s, grazed intermittently until about 1981, and then abandoned (Radulovich and Sollins, 1985). Efforts concentrated on a grass site, covered with *Olyra latifolia* L. lightly interspersed with fern (*Pteridium* spp.). A second nearby site, covered by 15-yr-old, mixed secondary forest, was added later in the study. All measurements were made between 1984 and 1986.

The soil, a member of the Helechal series of Oxic Dystropepts (Sancho and Mata, 1987), has formed from alluvially deposited volcanic materials. Soil pH in water (0–10-cm depth; 1:5 soil/solution ratio) averaged 4.5 (SE = 0.1, $n = 4$). Base cation levels are very low ($1.9\ \text{cmol/kg}$ at 0 to 20-cm depth; Sancho and Mata, 1987); and the CEC is highly pH dependent with a point of zero charge ca 4.0 at 0 to 15-cm depth (Sollins et al., in press). The clay size fraction, mainly kaolinite, goethite, and gibbsite, accounts for about 70% (wt) of field-moist samples after dispersion by sonication in sodium pyrophosphate; many of the silt and sand-size particles present before dispersion are actually aggregates (Strickland et al., 1988). The aggregation is very stable relative to less weathered soils; even after heavy rainfall, the soil crumbles easily into aggregates of a few mm diameter.

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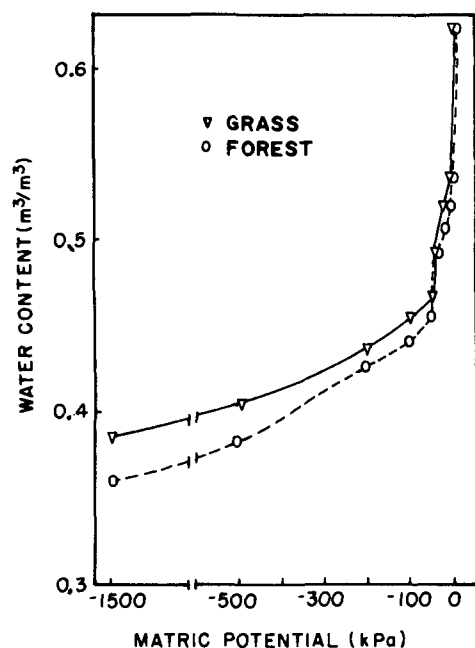


Fig. 1. Moisture release curves for soil samples from 10-cm depth under grass and forest. Each point is mean of four values ($CV < 4\%$ of mean). A bulk density of 0.8 Mg m^{-3} (weighted average for 10-cm depth) was used to convert from gravimetric to volumetric water content.

Bulk density, based on four soil pits in the grass area, averaged 0.73 Mg m^{-3} ($SE = 0.04$) at 0 to 10-cm depth and 0.87 Mg m^{-3} ($SE = 0.02$) from 10- to 100-cm depth. Particle density, measured with a pycnometer, averaged 2.53 Mg m^{-3} ($SE = 0.03$) at 0 to 10-cm depth and 2.67 ($SE = 0.02$) at 10 to 20-cm depth across both the grass and forest sites. Total porosity thus averaged 71% at 0 to 10-cm depth and 67% at 10 to 20-cm depth. Initial infiltration rate, with soil at field capacity, averaged 3900 mm hr^{-1} but was highly variable ($CV = 68\%$) (Radulovich and Sollins, 1985).

Field Methods

To study water drainage rates, duplicate plots (2.0 m^2) were established in grass and forest areas, and in a fern area within the grass site. Twice during the rainy season, when the soil was at or near field capacity, plots were cleared of vegetation and litter by hand without stepping on the exposed soil surface, in order to minimize compaction. After clearing, 40 L of water were applied on the surface in order to saturate the soil, then duplicate 10-cm deep soil cores were taken from each plot periodically for 10 d. To eliminate effects of evaporation and rainfall, the soil and surrounding vegetation were covered between sampling times with a plastic sheet (3 by 3 m). In addition, a trench was dug about 1-m uphill from each plot to preclude lateral water flow into it. Soil samples were oven-dried at 105°C for 48 h to determine water content.

Rhodamine B (saturated, $\sim 1:10 \text{ v/v}$) was used to trace water flow (Anderson and Bouma, 1973). At each of 9 locations (5 under grass, 4 under forest), a pit was dug to 1-m depth and an area next to the pit was cleared of vegetation and litter when the soil was at or near field capacity. About 5 L of dye solution was then applied as a shower from a watering can held 50-cm above the bare soil surface. The application rate ($\sim 0.05 \text{ mm s}^{-1}$), chosen to prevent ponding, simulated heavy rain. Immediately after applying the dye, the soil was removed at intervals of 2 cm, and the staining patterns at 10-, 20-, and 30-cm depth were traced on a transparent plastic sheet laid on the exposed surface. The tracings were then photographed.

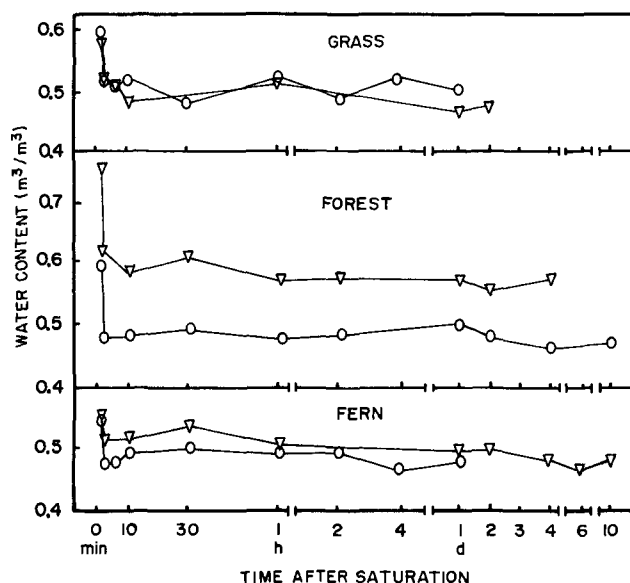


Fig. 2. Decrease in water content upon drainage after saturation. Each point is the mean of duplicate 10-cm deep cores ($CV < 4\%$ of mean). Different symbols distinguish measurements on different dates.

Samples for measurement of moisture release curves with pressure plates were obtained from 10- to 12-cm depth from the grass and forest sites and kept field moist; those subjected to pressures $\leq 200 \text{ kPa}$ were taken undisturbed in rings (Forsythe, 1980). Larger undisturbed soil cores, for breakthrough and dye-application experiments, were obtained by inserting wax-coated steel cylinders (10.75-cm i.d., 0.2-cm wall thickness) to a depth of 20 cm.

Laboratory Methods

The steady-state rate of water flow through 22 large cores was determined by manually adjusting the head over the soil surface until it was constant at 0.5 cm. Although referred to usually as "saturated" steady-state flow, we use the term ponded steady state flow to emphasize that the fine pores were not necessarily saturated (c.f., Beven and Germann, 1981). After the flow rate was measured, inflow was stopped, the soil core was allowed to drain, and then 1 L of the dye solution was added maintaining again a head of 0.5 cm. After the dye had percolated, the soil core was inverted, the lower 2 cm of soil removed, the stain pattern traced, and the total stained area measured. Total stained area was measured near the bottom of the cores because macroporosity decreased with depth.

A second set of 33 intact cores was used to obtain breakthrough curves (Nielsen and Biggar, 1961; 1962) for CaCl_2 . An initial concentration of 1 mM was chosen to match approximately the electrical conductivity of effluent from water-leached soil cores. Ponded steady-state flow through the cylinders was again maintained by keeping a 1-cm head above the soil. After electrical conductivity (EC) of the effluent had stabilized (termed reference EC), the concentration of the influent CaCl_2 solution was increased to 10 mM by allowing the 1-cm head of 1 mM CaCl_2 solution to drop to the soil surface, then bringing it immediately back up with the 10 mM CaCl_2 solution. During application of the initial weak CaCl_2 solution, effluent was sampled continuously and EC measured every 50 mL. After changing influent solution, EC was measured every 15 mL for the first 150 mL, then less frequently until a total of 5150 mL had passed through the core. Relative concentration of the effluent solution (C_r) was calculated as follows

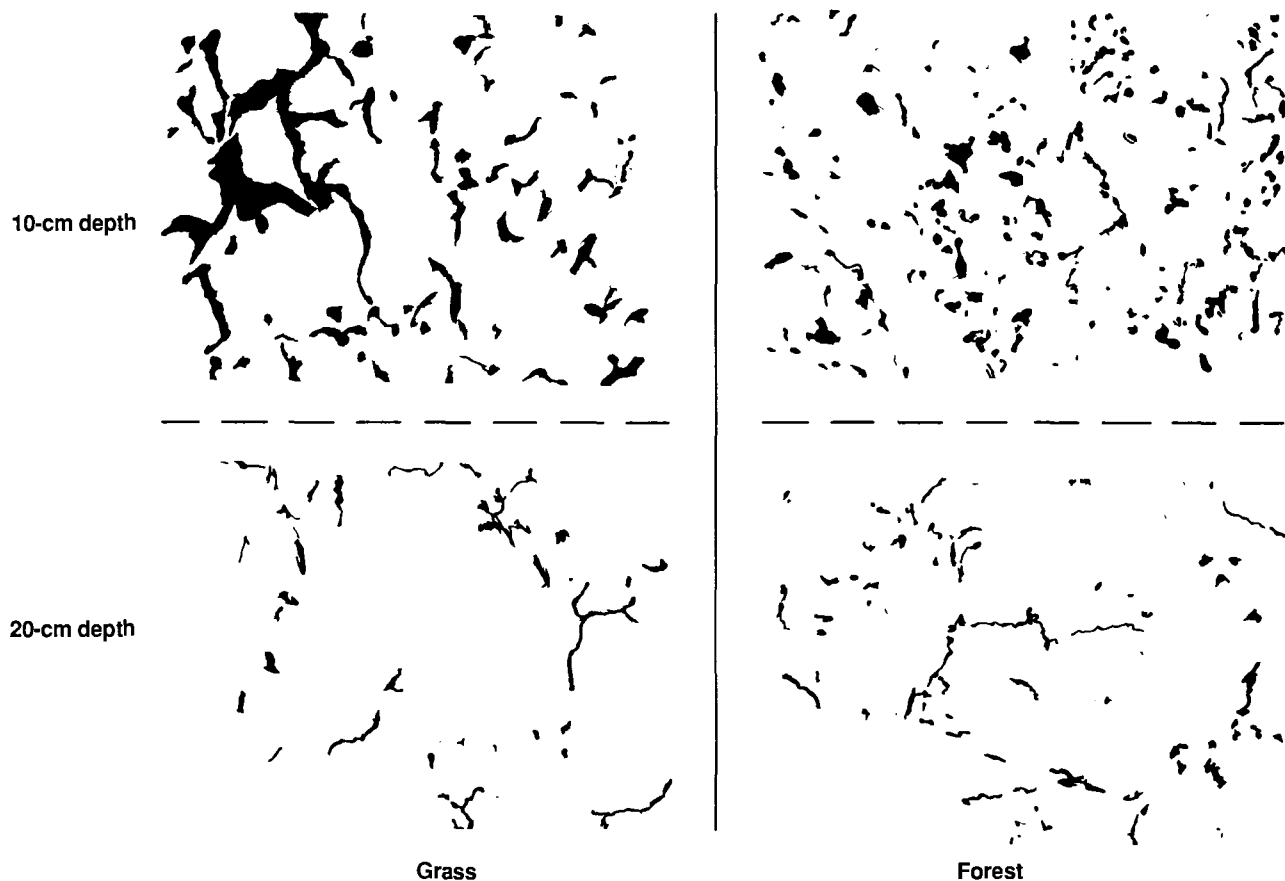


Fig. 3. Typical staining patterns resulting from field application of Rhodamine B. Patterns are for soil under grass and forest at 10- and 20-cm depth.

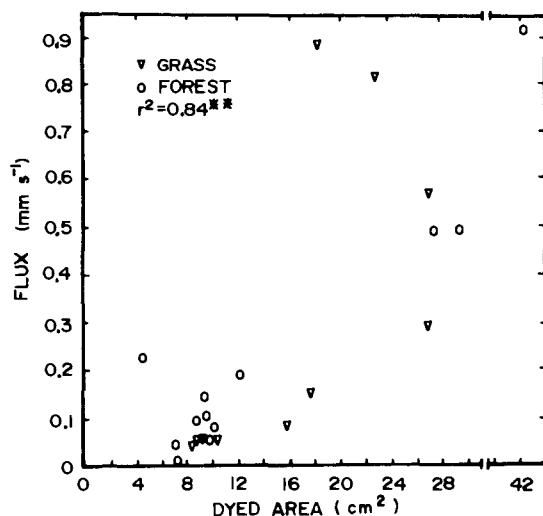


Fig. 4. Relation between ponded steady-state water flux and percent dyed area at 2-cm above the bottom of intact cores from under grass and forest. For forest cores, $r^2 = 0.91$ ($p < 0.01$); for grass cores, $r^2 = 0.81$ ($p < 0.05$).

$$C_r = (E_o - E_r) - (E_i - E_r)$$

where E_o , E_r , and E_i refer to electrical conductivity of the effluent, reference, and influent solutions, respectively. Interrupted breakthrough curves for five cores were obtained similarly except that inflow was stopped after about 1.8 pore volumes, the cylinders were allowed to drain for 10 min, and then inflow was resumed.

RESULTS

The La Selva soil released a large amount of water between saturation and field capacity, but relatively little between -50 and -1500 kPa (Fig. 1). The large amount of water (0.36 to 0.38 $\text{m}^3 \text{m}^{-3}$) remaining at -1500 kPa is reasonable given the high clay content of the La Selva soil. The difference in water content between saturation and field capacity (approx. -30 kPa) represents macroporosity—those pores that drain freely because they are too large to retain water by capillarity—and accounted for from 12.0 to 12.4% of the total pore space at the two sites.

Drainage slowed markedly within at most 10 min after saturation in all cases (Fig. 2), indicating that field capacity was reached quickly. Differences between the two curves for the forest soil could be due to higher organic-matter content in the soil that retained more water. All curves, however, show the extraordinary speed with which these soils drain to field capacity. Rapid drainage (Fig. 2), high infiltration rates (Radulovich and Sollins, 1985), the steep slope of the moisture release curves between 0 and -50 kPa (Fig. 1), and the large amount of water retained at -1500 kPa indicate that both fine pores (<0.1 μm diameter) and large pores (>30 μm) are abundant, which is typical of highly aggregated soils (Tsuji et al., 1975; Uehara and Gillman, 1981).

Dye patterns resulting from simulated heavy rainfall for grass and forest soil at two depths show that flow followed preferred paths along what appeared to

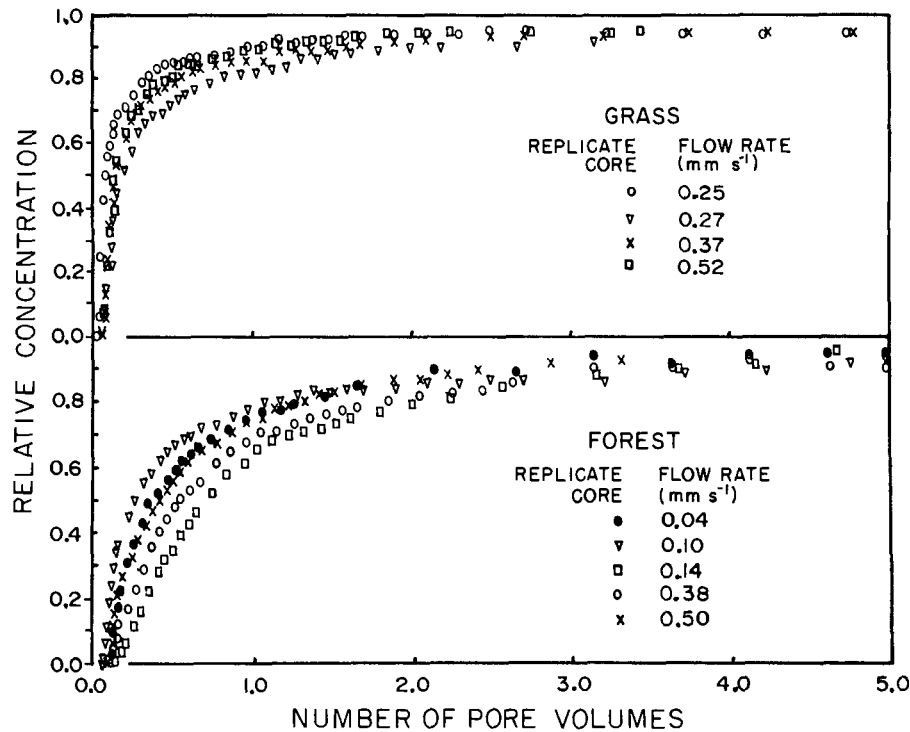


Fig. 5. Calcium chloride (CaCl_2) breakthrough curves for intact cores from grass and forest soil. Flow rate through each core was adjusted to steady-state.

be root channels, animal burrows, and cracks (e.g., Fig. 3). In general, macroporosity decreased with depth (c.f., McVoy, 1985). Heavily stained areas (e.g., at 10-cm depth under grass—Fig. 3) were common and were caused by spongy organic material, presumably decaying roots, rather than by voids. Dye experiments with intact cores gave similar staining patterns (not shown here). Ponded steady-state water flux through the cores varied markedly (from 0.01–0.91 mm s^{-1}) but was explained in large part by the variation in total macropore area (Fig. 4).

Calcium chloride (CaCl_2) breakthrough curves were similar for all the intact cores (Fig. 5), although ponded steady-state water flux ranged from 0.04 to 0.50 mm s^{-1} for the forest soil and from 0.25 to 0.52 mm s^{-1} for the grass soil. For the grass soil, especially, solute breakthrough occurred very early, often after outflow of less than 0.1 pore volume. Relative concentration then rose rapidly to about 0.7 for the forest soil and 0.8 for the grass soil. The rise slowed thereafter, and relative concentration failed to exceed 0.95 even after five pore volumes had percolated. When the inflow was stopped after about 1.8 pore volumes, then resumed after 10-min drainage, the relative concentration of the effluent dropped markedly in all cases (Fig. 6).

DISCUSSION

Most models of biphasic flow assume that the water in fine pores constitutes an immobile phase, whereas the water in large pores comprises a mobile phase. Solute flow between the two phases is assumed to be diffusion limited (e.g., Rao et al., 1982; Van Genuchten and Dalton, 1986). Biphasic flow can occur only in soils that are aggregated or contain macropores, be-

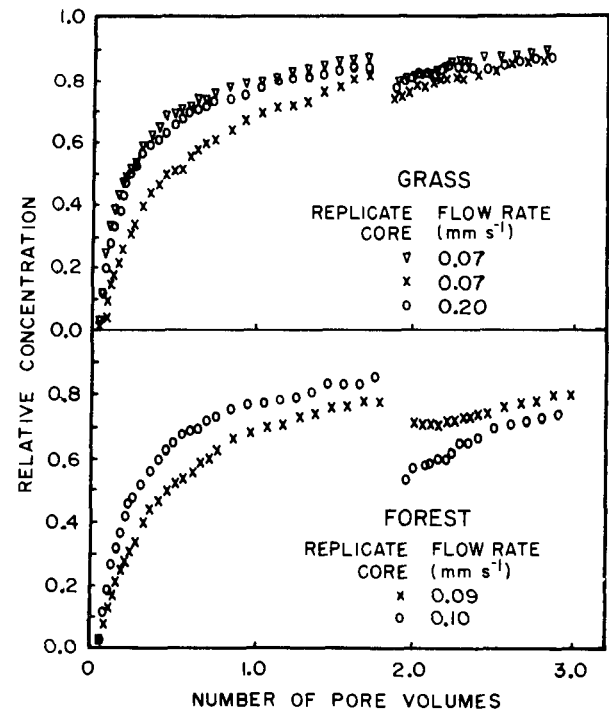


Fig. 6. Interrupted CaCl_2 breakthrough curves for intact cores taken from under grass and forest.

cause massive clay soils lack large pores, and sandy soils lack fine pores. That aggregate interiors need not be saturated for interaggregate flow to occur is well established; once the "infiltration capacity" of the aggregate exteriors or macropore walls is exceeded, a saturated zone builds up around (or over) which water

can flow at zero tension (Thomas and Phillips, 1979; Beven and Germann, 1981, 1982; White, 1985).

This biphasic conceptual model should apply well to the Oxic Dystropept at La Selva that we studied. The large amount of very fine (intra-aggregate) pore space is indicated by the high water content at -1500 kPa matric potential (Fig. 1) and by the resistance of the soil to solute entry (Fig. 5 and 6). In fact, calculations based on the moisture release curves show that pores <0.1 μm diameter account for more than 60% of the total porosity of this soil.

Extensive macroporosity is shown by the large and rapid changes in water content between saturation and field capacity (Fig. 1 and 2), the extremely high infiltration rates (Radulovich and Sollins, 1985), the dye patterns obtained in the field (Fig. 3), the high but variable rates of ponded steady-state flow in the laboratory (Fig. 4), and the early breakthrough of percolating CaCl_2 solution (Fig. 5). Admittedly, natural rainfall never creates a 1-cm head above the surface of this soil, so our experiments overstate somewhat the slowness of solute exchange between matrix and percolating solution. Nonetheless, relative to many agricultural and other less structured soils, in which a relative concentration of 0.95 is exceeded after as little as one pore volume has percolated (e.g., White, 1985), solute exchange in the La Selva soil is strikingly slow.

The decrease in relative concentration when steady-state flow is re-established after a pause (Fig. 6) can be explained as an effect of diffusion-limited exchange between the immobile intra- and the rapidly percolating interaggregate solutions. Thus the high relative concentrations reached at two to three pore volumes (Fig. 5) result from nearly complete saturation of aggregate exteriors. While flow was interrupted, CaCl_2 continued to diffuse into aggregates, thus lowering concentrations near the exterior of the aggregates. Immediately after flow was resumed, solute moved from the mobile to the immobile phase faster than it had immediately before flow was interrupted, and concentrations in the effluent dropped correspondingly. Many factors affect rates of solute diffusion within aggregates, including interactions with other dissolved species and electrostatic interactions at micropore surfaces (Schulin et al., 1986).

Preferential water flow, caused by aggregation and macropore development, may serve to prevent nutrient leaching in many ecosystems. Many tropical forests, for example, are developed on highly weathered, cation-poor soils (e.g., Oxisols), which tend to be highly aggregated simply because of their low charge (e.g., El-Swaify, 1980). Forest soils, in general, have well developed macropores because of animal activity and turnover of woody roots. Thus the leaching resistance resulting from preferential flow may help to explain not only the continued productivity of tropical forests on highly weathered soils, but also the marked decreases in productivity that have sometimes occurred upon their conversion to agriculture (Jordan, 1985; Ewel, 1986). For example, conversion from forest to pasture or annual crops eliminates woody roots whose continued growth and death is essential for macropore development. Subsequent cultivation, and perhaps liming, may decrease both aggregation and macroporosity even further. Loss of macroporosity and stable

aggregation is not entirely bad, however—a decrease in the amount of preferential flow should improve fertilizer application efficiency and reduce the potential for groundwater pollution (White, 1985).

The nutrient cycling implications of preferential flow may extend beyond the highly weathered soils of the humid tropics to other types of well aggregated soils (e.g., volcanic derived soils), and to forest soils and other noncultivated soils in which animal burrows and the channels from decaying roots are abundant. Even cultivated soils may have extensive macroporosity below the plough layer (see White, 1985).

Interestingly, over the course of soil development in the humid tropics, decreasing ion exchange capacity and increasing stability and extent of aggregation go hand in hand, both consequences of a decrease in total surface charge (Uehara and Gillman, 1981; Sollins et al., in press). Thus as weathering decreases a soil's capacity to retain ions on exchange sites, it may increase its capacity to resist leaching.

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